

MLFF CUEQ-DEFAULT1-HF1 Workstation Hotfix

Gate: CUEQ-DEFAULT1-HF1

Release: mdstats 0.20.194a0

Architecture revision: 61

This hotfix preserves the revision-60 split: source inference, DATA6, pseudolabels, evaluation, and verification use e3nn; TRAIN2 uses pure CuEq; exported checkpoints remain portable e3nn form (`only_cueq=false`).

Foundation-contract correction

FoundationConfigContract.v2 carries `source_backend` and `training_backend`. doctor now reports those exact fields and never indexes the retired aggregate backend field.

Numerical authority

The generic ACCEL1 FP32 source/DATA6 parity policy remains unchanged at `rtol=1e-5`, `atol=1e-6`.

The EXTRACT1-derived selected-head TRAIN2 starting model uses a separate FP32 preflight policy: `rtol=1e-5`, `atol=2e-6`. FP64 remains `rtol=1e-10`, `atol=1e-12` for both authorities.

The TRAIN2 floor is frozen from the earlier selected-head workstation experiment that motivated CUEQ-PHASE1 ($F_{\max}$ about $1.669e-6$, $D_{\max}$ about $1.192e-6$, identical deterministic selection). It is not derived from locked-test tuning and does not change source/DATA6 acceptance.

Workstation result disposition

The reported values $E_{\max}=3.576e-7$, $F_{\max}=1.490e-6$, $S_{\max}=1.564e-8$, $D_{\max}=1.570e-6$, with `selection_identical=True`, are inside the frozen TRAIN2 FP32 policy. They are intentionally not used to alter the generic source/DATA6 policy.

Fail-closed requirements

TRAIN2 CuEq remains non-authorizing if the CuEq/CUDA runtime freeze fails, the selected-head checkpoint identity is invalid, any energy/force/stress/descriptor output is non-finite, deterministic selection fingerprints differ, or numerical differences exceed the dedicated TRAIN2 policy. No silent e3nn fallback is allowed.